

Double/Debiased Machine Learning

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1 Recap

1.1 The Partially Linear Model

Machine-learning methods provide a flexible way to control for observed covariates without requiring the researcher to specify their exact functional form. A prominent example that combines an interpretable econometric model with flexible machine learning is the **partially linear model**:

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \quad \mathbb{E}[U_i \mid D_i, X_i] = 0,$$

where Y_i is the outcome, D_i is the variable of interest, and X_i is a vector of observed covariates. The scalar θ_0 is the target parameter, whereas $g_0(X_i)$ is an unknown nuisance function describing how the controls enter the outcome equation.

1.2 Nuisance Components and Moment Functions

1.2.1 With $\mathbb{E}[D_i \mid X_i] = 0$

Under the simplifying assumption

$$\mathbb{E}[D_i \mid X_i] = 0,$$

taking conditional expectations of the outcome equation gives

$$\mathbb{E}[Y_i \mid X_i] = g_0(X_i).$$

Hence, there is only one nuisance function to learn:

$$\eta_0 = g_0, \quad g_0(X) = \mathbb{E}[Y \mid X].$$

A machine-learning regression of Y on X therefore directly estimates the nuisance component η_0 .

Using this nuisance function, consider the moment function

$$m(W_i; \theta, \eta) = D_i\{Y_i - \theta D_i - \eta(X_i)\}, \quad W_i = (Y_i, D_i, X_i).$$

At the truth,

$$m(W_i; \theta_0, \eta_0) = D_i U_i,$$

and therefore

$$\mathbb{E}[m(W_i; \theta_0, \eta_0)] = \mathbb{E}[D_i U_i] = 0.$$

Thus, under $\mathbb{E}[D_i \mid X_i] = 0$, the target parameter satisfies

$$\mathbb{E}[D_i\{Y_i - \theta_0 D_i - g_0(X_i)\}] = 0.$$

1.2.2 Without $\mathbb{E}[D_i \mid X_i] = 0$

We now allow D_i to depend systematically on X_i . Define

$$d_0(X_i) := \mathbb{E}[D_i \mid X_i], \quad V_i := D_i - d_0(X_i),$$

so that

$$D_i = d_0(X_i) + V_i, \quad \mathbb{E}[V_i \mid X_i] = 0.$$

Taking conditional expectations of the outcome equation gives

$$h_0(X_i) := \mathbb{E}[Y_i \mid X_i] = \theta_0 d_0(X_i) + g_0(X_i).$$

Thus, a regression of Y on X no longer directly estimates g_0 . Instead, there are two prediction problems:

$$\begin{array}{ll} Y \text{ on } X & \longrightarrow h_0(X) = \mathbb{E}[Y \mid X], \\ D \text{ on } X & \longrightarrow d_0(X) = \mathbb{E}[D \mid X]. \end{array}$$

It is therefore convenient to collect the nuisance functions as

$$\eta_0 = (h_0, d_0).$$

The corresponding moment function is

$$m(W_i; \theta, \eta) = \{D_i - d(X_i)\} [Y_i - h(X_i) - \theta\{D_i - d(X_i)\}], \quad \eta = (h, d).$$

Why? Start from the structural error,

$$U_i = Y_i - \theta_0 D_i - g_0(X_i).$$

The part of D_i that remains after accounting for X_i is

$$V_i = D_i - d_0(X_i).$$

Because V_i is a function of (D_i, X_i) ,

$$\mathbb{E}[V_i \mid D_i, X_i] = V_i.$$

Together with the exogeneity condition

$$\mathbb{E}[U_i \mid D_i, X_i] = 0,$$

this gives

$$\begin{aligned} \mathbb{E}[V_i U_i] &= \mathbb{E}[\mathbb{E}[V_i U_i \mid D_i, X_i]] \\ &= \mathbb{E}[V_i \mathbb{E}[U_i \mid D_i, X_i]] \\ &= 0. \end{aligned}$$

Thus, $V_i U_i$ provides a valid population moment.

To express this moment using nuisance functions that can be learned directly from prediction problems, note that

$$g_0(X_i) = h_0(X_i) - \theta_0 d_0(X_i).$$

Substituting this into the structural error gives

$$\begin{aligned} U_i &= Y_i - \theta_0 D_i - g_0(X_i) \\ &= Y_i - h_0(X_i) - \theta_0 \{D_i - d_0(X_i)\}. \end{aligned}$$

Combining this expression with

$$V_i = D_i - d_0(X_i)$$

yields

$$V_i U_i = \{D_i - d_0(X_i)\} [Y_i - h_0(X_i) - \theta_0 \{D_i - d_0(X_i)\}].$$

This directly motivates the moment function above. At the true parameter and nuisance functions,

$$m(W_i; \theta_0, \eta_0) = V_i U_i,$$

and therefore

$$\mathbb{E}[m(W_i; \theta_0, \eta_0)] = \mathbb{E}[V_i U_i] = 0.$$

Hence, without imposing $\mathbb{E}[D_i \mid X_i] = 0$, the target parameter satisfies

$$\boxed{\mathbb{E}[\{D_i - d_0(X_i)\} \{Y_i - h_0(X_i) - \theta_0 [D_i - d_0(X_i)]\}] = 0.}$$

The earlier setup is recovered as the special case

$$d_0(X) = 0, \quad h_0(X) = g_0(X),$$

so the general formulation nests the simpler one.

1.3 Cross-Fitting

How nuisance error affects the target estimate. Let η_0 denote the nuisance component, and let $\hat{\eta}$ be its machine-learning estimate. Let $m(W_i; \theta, \eta)$ denote the moment function used to recover the target parameter θ_0 .

The target estimator $\hat{\theta}$ is chosen so that the sample moment is approximately zero. If $\hat{\eta}$ differs from η_0 , this nuisance-estimation error can shift the moment away from zero. The target estimate must then move to offset that shift.

Schematically,

$$\boxed{\text{nuisance error} \longrightarrow \text{error in the moment} \longrightarrow \text{error in } \hat{\theta}.}$$

The first-order effect of nuisance estimation on the moment takes the schematic form

$$\underbrace{\frac{\partial m}{\partial \eta}}_{\text{sensitivity of the moment to the nuisance}} \underbrace{(\hat{\eta} - \eta_0)}_{\text{error from learning the nuisance}} .$$

Accordingly, the resulting contribution to target-parameter error can be represented schematically as

$$\hat{\theta} - \theta_0 \approx - \left[\mathbb{E} \left(\frac{\partial m}{\partial \theta} \right) \right]^{-1} \left[\mathbb{E} \left(\frac{\partial m}{\partial \eta} \right) (\hat{\eta} - \eta_0) \right] + \text{sampling variation}.$$

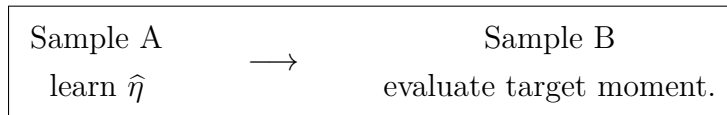
The derivative with respect to η measures how strongly nuisance error shifts the moment condition. The derivative with respect to θ determines how much θ must move to offset that shift.

What happens if the same sample is used twice? Suppose the same observations are used both to learn $\hat{\eta}$ and to estimate θ_0 . Then the sensitivity of the moment and the nuisance-estimation error are constructed from the same sample and can therefore be statistically related. Schematically,

$$\mathbb{E} \left[\frac{\partial m}{\partial \eta} (\hat{\eta} - \eta_0) \right] \neq \mathbb{E} \left[\frac{\partial m}{\partial \eta} \right] \mathbb{E} [\hat{\eta} - \eta_0] .$$

Thus, observation-specific irregularities that affect nuisance learning can also enter target-parameter estimation through the same observations.

Why separate the samples? Suppose instead that the nuisance component is learned using one sample and the target estimating equation is evaluated using another:



The error from learning the nuisance component is now generated from observations different from those used to evaluate the target estimating equation. Hence, observation-specific irregularities that affect nuisance learning do not simultaneously enter target estimation through the same observations.

This separation is the basic motivation for sample splitting:

use one subsample to learn $\hat{\eta}$,
use another subsample to evaluate the target moment.

From sample splitting to cross-fitting. A single sample split creates the desired separation, but it has an obvious cost: only part of the data is used for target-parameter estimation.

Cross-fitting keeps the separation while rotating the roles of the subsamples. For example, split the estimation sample into two nonoverlapping folds, A and B :

Fold A: estimate $\hat{\eta}^{(A)}$ $\longrightarrow$ Fold B: evaluate target moment
Fold B: estimate $\hat{\eta}^{(B)}$ $\longrightarrow$ Fold A: evaluate target moment.

Thus, every observation can contribute to target-parameter estimation, while the nuisance component applied to that observation is learned from a different subsample:

sample splitting gives separation + cross-fitting reuses the data efficiently.

In the general partially linear model,

$$\eta_0 = (h_0, d_0), \quad h_0(X) = \mathbb{E}[Y \mid X], \quad d_0(X) = \mathbb{E}[D \mid X].$$

Thus, when fold A is used for nuisance estimation, we fit

$$\hat{h}^{(A)}(X) \approx \mathbb{E}[Y \mid X], \quad \hat{d}^{(A)}(X) \approx \mathbb{E}[D \mid X],$$

using only observations in fold A . These fitted nuisance functions are then applied to observations in fold B .

The roles of the two folds are subsequently reversed. After every observation has received out-of-fold nuisance estimates, the observations are pooled and the target parameter is estimated using the cross-fitted moment condition.

Why cross-fitting alone is not enough. Cross-fitting removes the direct same-sample dependence between nuisance learning and evaluation of the target

estimating equation. However, it does not make the nuisance-estimation error small, nor does it generally make the sensitivity

$$\mathbb{E} \left[\frac{\partial m}{\partial \eta} \right]$$

equal to zero.

Thus, even after cross-fitting, nuisance-estimation error can still enter $\hat{\theta} - \theta_0$ at first order.

Suppose the usual sampling variation of the target estimator is

$$O_p(n^{-1/2}),$$

while a flexible machine-learning estimator of the nuisance component satisfies, for illustration,

$$\hat{\eta} - \eta_0 = O_p(n^{-1/4}).$$

If

$$\mathbb{E} \left[\frac{\partial m}{\partial \eta} \right] \neq 0,$$

then the nuisance contribution to target-parameter error can also be of order

$$O_p(n^{-1/4}).$$

Hence, schematically,

$$\hat{\theta} - \theta_0 = \underbrace{O_p(n^{-1/2})}_{\text{sampling variation}} + \underbrace{O_p(n^{-1/4})}_{\text{nuisance contribution}}.$$

Both terms converge to zero, but at very different rates:

$$n^{-1/4} \gg n^{-1/2}.$$

For example,

$$10,000^{-1/4} = 0.10, \quad 10,000^{-1/2} = 0.01.$$

Thus, nuisance-estimation error can remain much larger than the sampling variation relevant for inference on θ_0 . Consistency of $\hat{\eta}$ alone is therefore not enough.

Nuisance consistency is not enough:
the error may vanish too slowly for root- n inference.

What do we need in addition to cross-fitting? Because nuisance-estimation error itself may be non-negligible, we want the estimating equation to have zero first-order sensitivity to that error:

$$\mathbb{E} \left[\frac{\partial m}{\partial \eta} \right] = 0.$$

If this sensitivity is zero, nuisance-estimation error does not shift the moment condition at first order and therefore does not enter $\hat{\theta} - \theta_0$ at first order. This is the central motivation for **Neyman orthogonalization**.

Cross-fitting separates nuisance learning from target estimation,
Neyman orthogonality blocks the first-order transmission
of nuisance error into target-parameter error.

2 Neyman Orthogonality

The purpose of Neyman orthogonality is to make estimation of the target parameter *insensitive* to errors in the nuisance component¹.

A useful way to understand this idea is through **residualization**. The following geometric picture is an analogy: the target estimating component may contain a part associated with nuisance variation. We remove that part and retain the residualized component. The remaining component represents the part of the target estimating equation that is locally insensitive to nuisance variation.

¹More precisely, the insensitivity is local. “Locally insensitive” means that, near the true nuisance component, a sufficiently small estimation error in the nuisance does not affect the population estimating equation at first order. Equivalently, the leading linear effect of nuisance error is removed, so any remaining effect enters through higher-order terms. This is a local robustness property and does not generally imply statistical independence between the target and nuisance components.

2.1 Geometric Derivation

Step 1: Draw the nuisance space. Begin with the nuisance space, represented by a plane. The plane represents directions in which the nuisance component can vary.



Figure 1: Step 1: the nuisance space represents directions associated with variation in the nuisance component.

Step 2: Add the target estimating component. Now add the original target estimating component T_θ . Before orthogonalization, it may contain a part associated with nuisance variation.

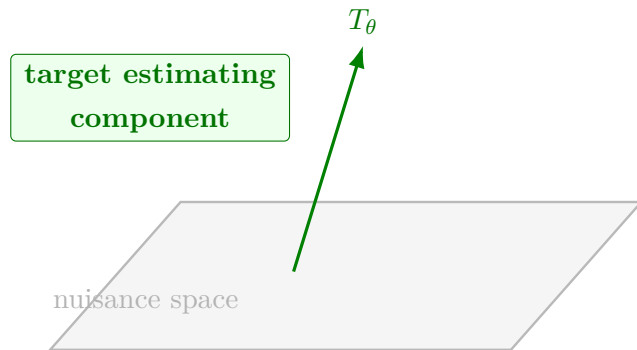


Figure 2: Step 2: the original target estimating component T_θ .

Step 3: Identify the nuisance-related part. The target estimating component may contain a part that lies in the nuisance space. Denote this part by $T_{\theta|\eta}$. It represents the component of the target estimating equation associated with nuisance variation.

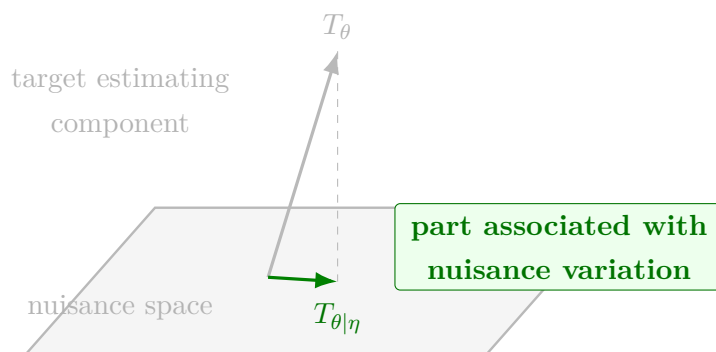


Figure 3: Step 3: $T_{\theta|\eta}$ represents the nuisance-related part of the target estimating component.

Step 4: Remove the nuisance-related part. Subtract the nuisance-related component from the original target component. The remaining part, denoted $\tilde{T}_\theta$, is the residualized target component.

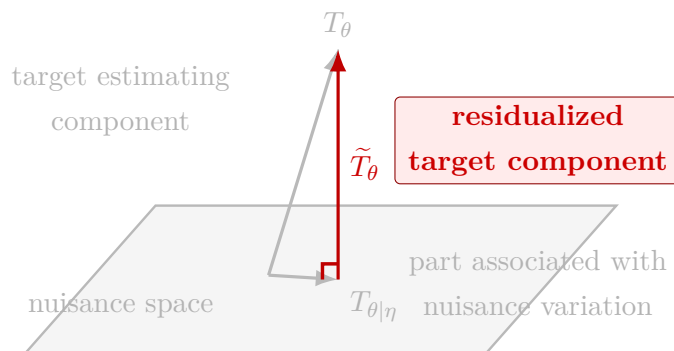


Figure 4: Step 4: removing the nuisance-related component leaves the residualized target component $\tilde{T}_\theta$. The right angle represents orthogonality.

Why residualization protects target estimation. The residualized target component is protected from nuisance-estimation error because the nuisance-related direction has been removed. This is the core intuition behind Neyman orthogonality.²

²More precisely, this protection is local and first-order: for sufficiently small perturbations around the true nuisance component, the leading linear effect on the population estimating equation is zero. Any remaining effect enters through higher-order terms.

2.2 Algebraic Derivation: From the Moment Condition to Residualized Regression

Recall the population moment condition introduced earlier:

$$\mathbb{E} [\{D_i - d_0(X_i)\} \{Y_i - h_0(X_i) - \theta_0[D_i - d_0(X_i)]\}] = 0.$$

Define

$$\tilde{Y}_i = Y_i - h_0(X_i), \quad \tilde{D}_i = D_i - d_0(X_i).$$

Then the moment condition becomes

$$\mathbb{E} [\tilde{D}_i (\tilde{Y}_i - \theta_0 \tilde{D}_i)] = 0.$$

Expanding,

$$\mathbb{E}[\tilde{D}_i \tilde{Y}_i] - \theta_0 \mathbb{E}[\tilde{D}_i^2] = 0.$$

Provided that

$$\mathbb{E}[\tilde{D}_i^2] > 0,$$

the target parameter is therefore

$$\boxed{\theta_0 = \frac{\mathbb{E}[\tilde{D}_i \tilde{Y}_i]}{\mathbb{E}[\tilde{D}_i^2]}.$$

Replacing the population moments by their sample analogues gives

$$\frac{1}{n} \sum_{i=1}^n \tilde{D}_i (\tilde{Y}_i - \theta \tilde{D}_i) = 0,$$

which solves to

$$\boxed{\hat{\theta} = \frac{\sum_{i=1}^n \tilde{D}_i \tilde{Y}_i}{\sum_{i=1}^n \tilde{D}_i^2}.$$

This is exactly the OLS coefficient from the residual-on-residual regression through the origin:

$$\boxed{\tilde{Y}_i = \theta \tilde{D}_i + \text{residual}_i.}$$

Thus, the residualized-regression estimator follows algebraically from the moment condition derived earlier.

Connection to the geometry. This estimator has the geometric interpretation developed in Steps 1–4 of the previous subsection. That is, the nuisance-related component is projected out of the target estimating component, leaving the residualized part. Algebraically,

$$\tilde{Y}_i = Y_i - h_0(X_i), \quad \tilde{D}_i = D_i - d_0(X_i).$$

The residual-on-residual regression therefore uses the variation in Y and D that remains after the nuisance components have been removed.

3 Double/Debiased Machine Learning

Double/debiased machine learning (DML) combines the two adjustments developed above. Both are designed to protect estimation of the target parameter from errors introduced when flexible machine-learning methods are used to estimate nuisance components.

- 1. Neyman orthogonality** : remove first-order sensitivity to nuisance error,
- 2. Cross-fitting** : separate nuisance learning from target estimation.

These two adjustments address different problems. Neyman orthogonality changes the estimating equation so that small nuisance-estimation errors do not affect it at first order. Cross-fitting changes how the data are used: the nuisance functions applied to an observation are learned from other observations. Together, they allow flexible machine-learning methods to be used for nuisance estimation while preserving reliable estimation and inference for the low-dimensional target parameter θ_0 .

3.1 How DML Works in Practice

Recall the partially linear model

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i,$$

with nuisance functions

$$h_0(X) = \mathbb{E}[Y \mid X], \quad d_0(X) = \mathbb{E}[D \mid X].$$

The DML procedure combines cross-fitting with residualization.

Step 1: Learn the nuisance components using cross-fitting. Divide the estimation sample into folds. For an observation in fold k , estimate the nuisance functions using observations outside that fold:

$$\hat{h}^{(-k)}(X) \approx \mathbb{E}[Y \mid X], \quad \hat{d}^{(-k)}(X) \approx \mathbb{E}[D \mid X].$$

Thus, for $i \in I_k$, define

$$\hat{h}_i = \hat{h}^{(-k)}(X_i), \quad \hat{d}_i = \hat{d}^{(-k)}(X_i).$$

Each observation therefore receives nuisance predictions from models that were trained without its own fold.

Step 2: Residualize with respect to the nuisance components. Construct

$$\tilde{Y}_i = Y_i - \hat{h}_i, \quad \tilde{D}_i = D_i - \hat{d}_i.$$

The residualized variables remove the components of Y_i and D_i predicted by the nuisance functions:

$\begin{aligned} Y_i &\longrightarrow \tilde{Y}_i = Y_i - \hat{h}_i, \\ D_i &\longrightarrow \tilde{D}_i = D_i - \hat{d}_i. \end{aligned}$
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Thus, $\tilde{Y}_i$ is the variation in the outcome remaining after removing the part explained by X_i , while $\tilde{D}_i$ is the variation in D_i remaining after removing the part explained by X_i .

This residualization is the practical manifestation of the Neyman-orthogonality idea developed in the previous section.

Step 3: Estimate the target parameter from the residualized variation. After every observation has received cross-fitted nuisance predictions, pool the residualized observations and run

$$\tilde{Y}_i = \theta \tilde{D}_i + \text{residual}_i.$$

Equivalently, solve the moment condition

$$\frac{1}{n} \sum_{i=1}^n \widetilde{D}_i (\widetilde{Y}_i - \theta \widetilde{D}_i) = 0.$$

Solving this equation gives

$$\widehat{\theta} = \frac{\sum_{i=1}^n \widetilde{D}_i \widetilde{Y}_i}{\sum_{i=1}^n \widetilde{D}_i^2}.$$

Hence, DML can be summarized as

use cross-fitting to learn the nuisance components
 $\Downarrow$
 residualize Y and D with respect to those components
 $\Downarrow$
 estimate θ_0 using the residualized variation.

The two debiasing devices enter at different stages:

Cross-fitting determines *which observations* learn the nuisance,
 Neyman orthogonality determines *how nuisance error* enters the target equation.

3.2 Pseudo-Algorithm for Two-Fold DML

The two-fold DML procedure alternates the roles of the two folds so that every observation receives nuisance predictions from a model trained on a different subsample. After the rotations are complete, the resulting residuals are pooled and used to estimate the target parameter.

Rotation 1: train on Fold B, residualize Fold A. Use only observations in Fold B to estimate the nuisance functions

$$\eta_0 = (h_0, d_0), \quad h_0(X) = \mathbb{E}[Y \mid X], \quad d_0(X) = \mathbb{E}[D \mid X].$$

Denote the fitted nuisance functions by

$$\widehat{h}^{(-A)}(X), \quad \widehat{d}^{(-A)}(X),$$

where the superscript $(-A)$ indicates that Fold A was excluded from nuisance estimation.

For every observation $i \in A$, form the out-of-fold residuals

$$\tilde{Y}_i = Y_i - \hat{h}^{(-A)}(X_i), \quad \tilde{D}_i = D_i - \hat{d}^{(-A)}(X_i).$$

Rotation 2: train on Fold A, residualize Fold B. Reverse the roles of the folds. Use only observations in Fold A to estimate

$$\hat{h}^{(-B)}(X), \quad \hat{d}^{(-B)}(X).$$

For every observation $i \in B$, form

$$\tilde{Y}_i = Y_i - \hat{h}^{(-B)}(X_i), \quad \tilde{D}_i = D_i - \hat{d}^{(-B)}(X_i).$$

After the two rotations, every observation has exactly one pair of out-of-fold residuals:

$$(\tilde{Y}_i, \tilde{D}_i).$$

Pool the residuals and estimate the target parameter. Combine the residuals from Folds A and B and solve

$$\frac{1}{n} \sum_{i=1}^n \tilde{D}_i (\tilde{Y}_i - \theta \tilde{D}_i) = 0.$$

Equivalently, run the residual-on-residual regression through the origin:

$$\hat{\theta} = \frac{\sum_{i=1}^n \tilde{D}_i \tilde{Y}_i}{\sum_{i=1}^n \tilde{D}_i^2}.$$

The full two-fold procedure can therefore be summarized as

Rotation 1: B learns $\hat{\eta}^{(-A)} \longrightarrow A$ receives $(\tilde{Y}, \tilde{D})$

Rotation 2: A learns $\hat{\eta}^{(-B)} \longrightarrow B$ receives $(\tilde{Y}, \tilde{D})$

$\Downarrow$

pool all out-of-fold residuals

$\Downarrow$

estimate one final $\hat{\theta}$.

More generally, with K folds, repeat the same procedure for each fold: estimate the nuisance functions using all observations outside the held-out fold, construct residuals for the held-out observations, and then pool all cross-fitted residuals to estimate θ_0 .

3.3 Numerical Example

Consider the simple model

$$D_i = 10X_i + V_i, \quad Y_i = 2D_i + 4 - 6X_i,$$

where

$$\mathbb{E}[V_i \mid X_i] = 0.$$

For this illustration, take $U_i = 0$. The true target parameter is therefore

$$\theta_0 = 2.$$

The nuisance functions are

$$d_0(X) = \mathbb{E}[D \mid X] = 10X$$

and

$$\begin{aligned} h_0(X) &= \mathbb{E}[Y \mid X] \\ &= 2d_0(X) + 4 - 6X \\ &= 4 + 14X. \end{aligned}$$

Thus,

$$\boxed{d_0(X) = 10X, \quad h_0(X) = 4 + 14X.}$$

Consider the following estimation sample:

i	X_i	D_i	Y_i	Fold
1	0	1	6	A
2	0	1	6	B
3	0	-1	2	A
4	0	-1	2	B
5	1	11	20	A
6	1	11	20	B
7	1	9	16	A
8	1	9	16	B

Hence,

$$I_A = \{1, 3, 5, 7\}, \quad I_B = \{2, 4, 6, 8\}.$$

3.3.1 First debiasing device: cross-fitting

To construct nuisance estimates for observations in fold A , use only fold B .

Suppose we fit one prediction model for D given X and another for Y given X . Because fold B contains one high and one low value of D for each value of X , the fitted conditional means are

$$\begin{aligned} \hat{d}^{(-A)}(0) &= \frac{1 + (-1)}{2} = 0, & \hat{d}^{(-A)}(1) &= \frac{11 + 9}{2} = 10, \\ \hat{h}^{(-A)}(0) &= \frac{6 + 2}{2} = 4, & \hat{h}^{(-A)}(1) &= \frac{20 + 16}{2} = 18. \end{aligned}$$

These nuisance functions are then applied to observations in fold A .

The roles of the folds are subsequently reversed:

$B : \hat{\eta}^{(-A)} \longrightarrow A : \text{construct residuals},$
$A : \hat{\eta}^{(-B)} \longrightarrow B : \text{construct residuals}.$

Thus, no observation is used to construct the nuisance predictions applied to its own fold.

3.3.2 Second debiasing device: Neyman orthogonality through residualization

For observations in fold A , construct

$$\widetilde{D}_i = D_i - \widehat{d}^{(-A)}(X_i), \quad \widetilde{Y}_i = Y_i - \widehat{h}^{(-A)}(X_i).$$

This gives

i	D_i	$\widehat{d}_i$	$\widetilde{D}_i$	Y_i	$\widehat{h}_i$	$\widetilde{Y}_i$
1	1	0	1	6	4	2
3	-1	0	-1	2	4	-2
5	11	10	1	20	18	2
7	9	10	-1	16	18	-2

For example, observation 5 has

$$X_5 = 1, \quad D_5 = 11, \quad Y_5 = 20.$$

Its nuisance predictions, learned using fold B , are

$$\widehat{d}_5 = 10, \quad \widehat{h}_5 = 18.$$

Therefore,

$$\widetilde{D}_5 = 11 - 10 = 1,$$

and

$$\widetilde{Y}_5 = 20 - 18 = 2.$$

The large original values $D_5 = 11$ and $Y_5 = 20$ contain substantial variation explained by $X_5 = 1$. Residualization removes this nuisance-related part and retains only

$$(\widetilde{D}_5, \widetilde{Y}_5) = (1, 2).$$

This is precisely the intuition of the geometric picture from the previous section:

remove the nuisance-related component $\longrightarrow$ retain the residualized variation.

Because fold A has the same balanced structure as fold B , reversing the folds produces the same residual pairs for observations $(2, 4, 6, 8)$:

$$(1, 2), \quad (-1, -2), \quad (1, 2), \quad (-1, -2).$$

3.3.3 Pool the cross-fitted residuals

After rotating the folds, every observation has one pair of cross-fitted residuals. Across all eight observations,

$$\widetilde{D}_i \in \{-1, 1\}, \quad \widetilde{Y}_i = 2\widetilde{D}_i.$$

Therefore,

$$\sum_{i=1}^8 \widetilde{D}_i \widetilde{Y}_i = 16,$$

while

$$\sum_{i=1}^8 \widetilde{D}_i^2 = 8.$$

The DML estimator is therefore

$$\widehat{\theta} = \frac{16}{8} = 2 = \theta_0.$$

The example makes the two roles of DML transparent:

Cross-fitting:	the nuisance prediction for an observation is learned elsewhere,
Neyman orthogonality:	the nuisance-predicted parts of Y and D are removed.

After these two adjustments, θ_0 is estimated from the relationship between the variation in Y and the variation in D that remains after the nuisance components have been removed.

4 Economic Applications

A useful way to organize economic applications is to combine a structured economic component with a flexible machine-learning component:

$$Y = \underbrace{\theta_0 D}_{\text{model-based component}} + \underbrace{g_0(X)}_{\text{flexible ML component}} + \underbrace{U}_{\text{unobserved component}}.$$

DML is primarily a framework for **estimation and inference**. Machine learning is used to estimate nuisance components flexibly, while the main objective is valid inference for a low-dimensional economic parameter, such as a treatment effect, elasticity, or policy coefficient. DML is especially useful when θ_0 has a clear economic interpretation but the relationships involving the control variables are complex and are better learned flexibly.

Identification still matters. Machine learning does not by itself solve endogeneity. The research design must still justify the identifying assumptions. If D remains endogenous after conditioning on X , a different design, such as IV-DML, may be required.

Examples.

- **Monetary or fiscal policy effects:** estimate the effect of a policy shock on output, inflation, employment, or investment, while flexibly controlling for macroeconomic and financial conditions.
- **Trade exposure:** estimate the effect of import competition, export exposure, or a trade-policy change on employment, wages, or productivity, while flexibly adjusting for industry and regional characteristics.
- **Firm decisions:** estimate the effect of technology adoption, investment, or management practices on productivity or output, with flexible controls for firm, industry, and market characteristics.

DML for time-series data. DML can also be extended to time-series settings under suitable conditions on temporal dependence and with an appropriate sample-splitting scheme. We will revisit these extensions after introducing autoregressive (AR) and moving-average (MA) models.